

Levels of Measurement

Measuring Variables

- The process of measuring a variable requires a set of categories called a **scale of measurement** and a process that classifies each individual into one category.

Why Is Level of Measurement Important?

- Helps you decide what statistical analysis is appropriate on the values that were assigned
- Helps you decide how to interpret the data from that variable

The Levels of Measurement

- Nominal
- Ordinal
- Interval
- Ratio

Types of Measurement Scales

1. A **nominal scale** is an unordered set of categories identified only by name. Nominal measurements only permit you to determine whether two individuals are the same or different.
2. An **ordinal scale** is an ordered set of categories. Ordinal measurements tell you the direction of difference between two individuals.

Types of Measurement Scales

3. An **interval scale** is an ordered series of equal-sized categories. Interval measurements identify the direction and magnitude of a difference. The zero point is located arbitrarily on an interval scale.
4. A **ratio scale** is an interval scale where a value of zero indicates none of the variable. Ratio measurements identify the direction and magnitude of differences and allow ratio comparisons of measurements.

Nominal Measurement

- The values “name” the attribute uniquely.
- The value does not imply any ordering of the cases, for example, jersey numbers in football.
- Even though player 32 has higher number than player 19, you can't say from the data that he's greater than or more than the other.



Nominal

- **Examples:**
- In a survey, the variable 'sex' is a nominal scale of measurement because there are two possibilities, male and female, which cover the entire population under study.
- In a medical test, a lab animal may be either dead or alive. Every animal under study is in one of the two states and there is no animal that cannot be described by these two states.
- In Quantum mechanics, the measured spin of an electron is either $+1/2$ or $-1/2$. The measurement cannot yield any other value and this is true for any electron under study.

Ordinal Measurement

When attributes can be rank-ordered...

- Distances between attributes do not have any meaning, for example, code Educational Attainment as 0=less than H.S.; 1=some H.S.; 2=H.S. degree; 3=some college; 4=college degree; 5=post college

Is the distance from 0 to 1 the same as 3 to 4?



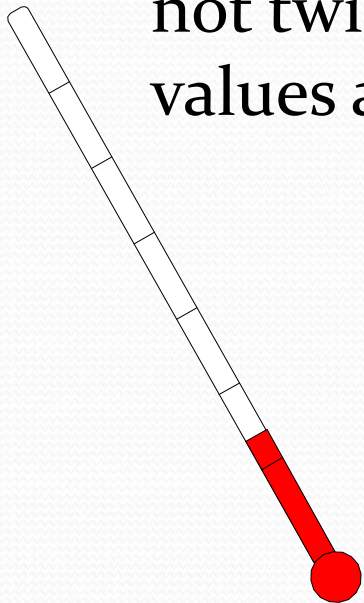
Ordinal

- **Examples:**
- In an Olympic race, the participants are ranked according to the ascending order of the time taken to finish the race. In this, the number tells us something about the relative performance of an athlete.
- The grading system used in university is a measurement scale of the type ordinal because there is a hierarchy involved.

Interval Measurement

When distance between attributes has meaning, for example, temperature (in Fahrenheit) -- distance from 30-40 is same as distance from 70-80.

- Note that ratios don't make any sense -- 80 degrees is not twice as hot as 40 degrees (although the attribute values are).



Interval

- **Examples:**
- In temperature measurement, we use scales that are interval measurement scales. The scales are also uniform in that the difference between 200C and 400C is the same as the difference between 400C and 600C.
- The number system that we use is another example of a uniform measurement scale.
- Not all interval measurement scales are uniform. A log scale is commonly used for plotting data, which is not uniform in nature.

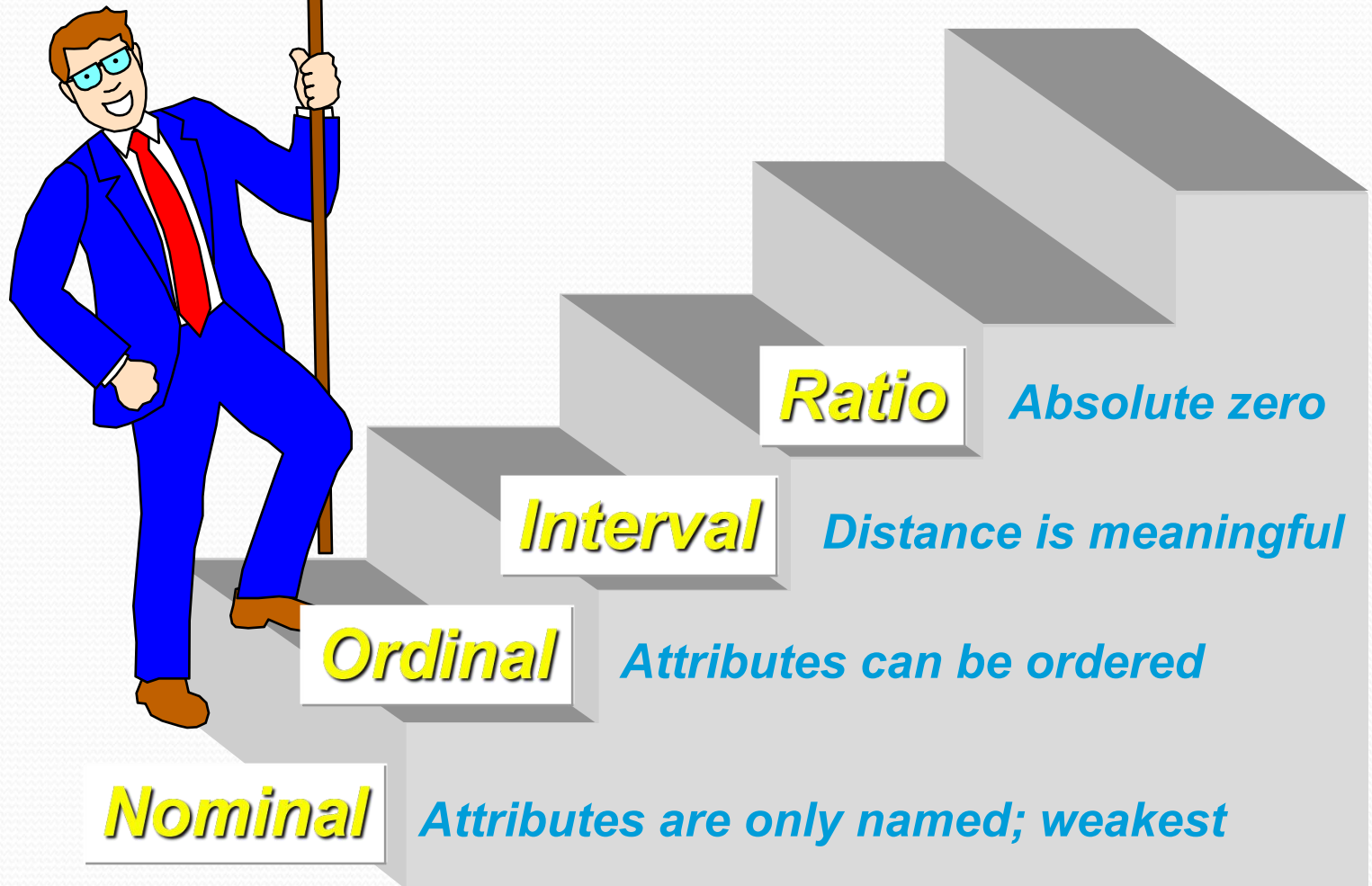
Ratio Measurement

- Has an absolute zero that is meaningful
- Can construct a meaningful ratio (fraction), for example, number of clients in past six months
- It is meaningful to say that “...we had twice as many clients in this period as we did in the previous six months.

Ratio

- An example of a ratio scale is the amount of money you have in your pocket right now (25 cents, 55 cents, etc.). Money is measured on a ratio scale because, in addition to having the properties of an interval scale, it has a true zero point: if you have zero money, this implies the absence of money. Since money has a true zero point, it makes sense to say that someone with 50 cents has twice as much money as someone with 25 cents (or that Bill Gates has a million times more money than you do).
- Most measurement in the physical sciences and engineering is done on ratio scales. Examples include mass, length, duration, plane angle, energy and electric charge.

The Hierarchy of Levels



Primary Scales of Measurement

Scale

<i>Nominal</i>	Numbers Assigned to Drivers/Cars	1	31	8
<i>Ordinal</i>	Rank Order of race finishers	Third Place	Second Place	First Place
<i>Interval</i>	Championship Points earned	170	175	185
<i>Ratio</i>	Time to Finish, behind winner	5.1	2.3	0.0

A Simple Example

- Six athletes try out for a sprinter's position on a local track team.
- They all run a 100 meter dash, and are timed by several coaches each using a different stopwatch.

A Simple Example (Nominal)

Athlete	True Time	V Nominal				
A	10	23				
B	11	12				
C	13	20				
D	20	19				
E	13	20				
S	0	26				

A Simple Example (Nominal)

- # Watch V is virtually useless, but it *has* captured a basic property of the running times. Namely, two values given by the watch are the same if and only if two actual times are the same.
- Watch V has achieved only a *nominal level of measurement*.

A Simple Example (Ordinal)

Athlete	True Time	V Nominal	W Ordinal			
A	10	23	11			
B	11	12	14			
C	13	20	15			
D	20	19	18			
E	13	20	15			
S	0	26	9			

A Simple Example (Ordinal)

- Besides capturing the same-difference property, Stopwatch W has the correct *ordering*.
- We say that Stopwatch W has achieved an *ordinal level of measurement*.

A Simple Example (Ordinal)

Athlete	True Time	V Nominal	W Ordinal	X Ordinal		
A	10	23	11	2		
B	11	12	14	3		
C	13	20	15	4		
D	20	19	18	5		
E	13	20	15	4		
S	0	26	9	1		

A Simple Example (Ordinal)

- Stopwatch X is also at the ordinal level of measurement!
- What does this tell you?

A Simple Example (Interval)

Athlete	True Time	V Nominal	W Ordinal	X Ordinal	Y Interval	
A	10	23	11	2	21	
B	11	12	14	3	23	
C	13	20	15	4	27	
D	20	19	18	5	41	
E	13	20	15	4	27	
S	0	26	9	1	1	

A Simple Example (Interval)

- The relative spacing (not the absolute spacing) of the values given by stopwatch Y matches the relative spacing of the actual times. So the intervals are in correct proportion.
- When the numbers capture same-difference, have the correct order, and have the correct relative interval spacing, we say they have achieved an interval level of measurement.

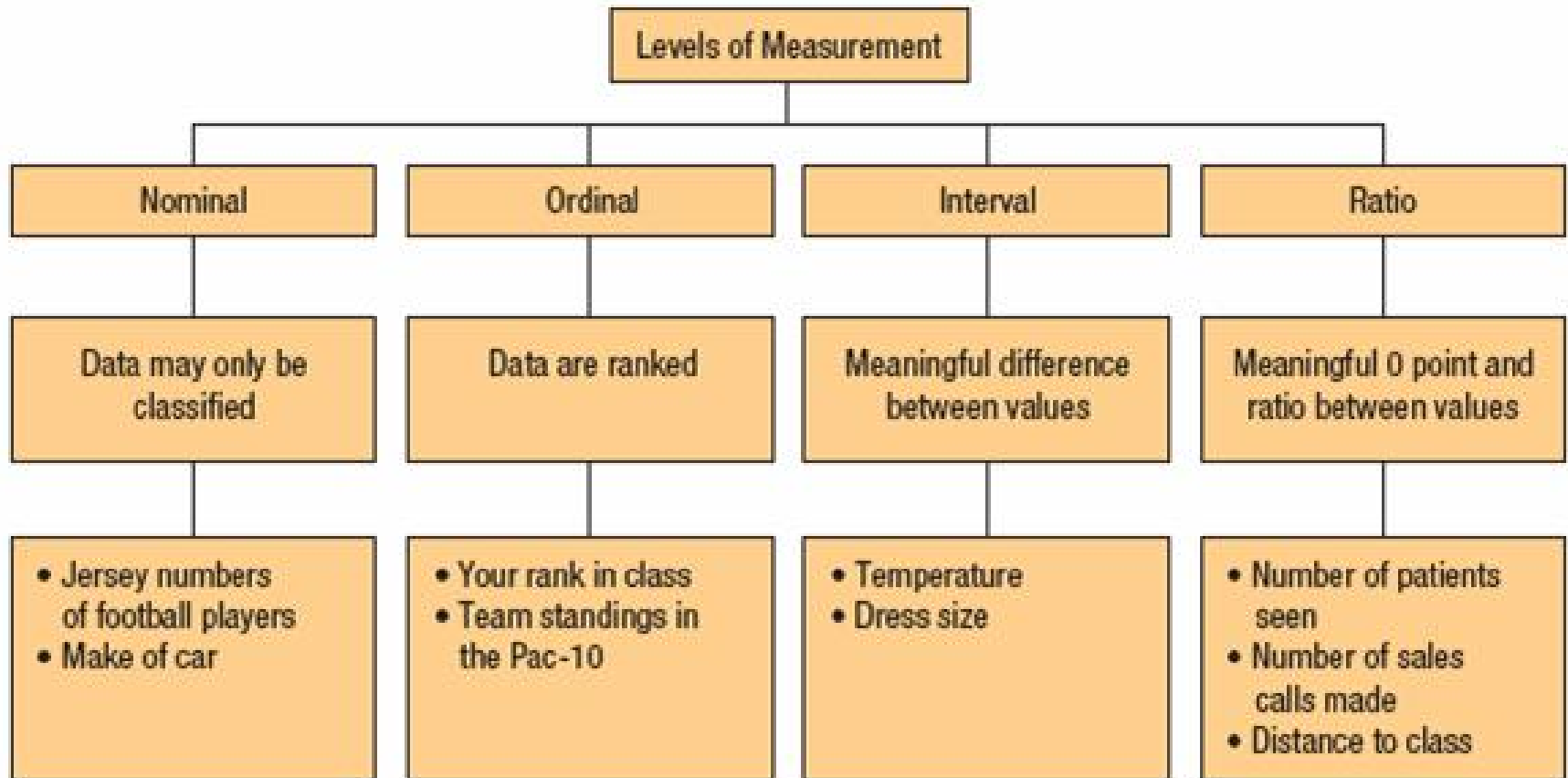
A Simple Example (Ratio)

Athlete	True Time	V Nominal	W Ordinal	X Ordinal	Y Interval	Z Ratio
A	10	23	11	2	21	20
B	11	12	14	3	23	22
C	13	20	15	4	27	26
D	20	19	18	5	41	40
E	13	20	15	4	27	26
S	0	26	9	1	1	0

A Simple Example (Ratio)

- The data produced by stopwatch Y do not capture ratios correctly. Person D took twice as long as person A, but the stopwatch did not assign a value that was twice as large.
- The zero point for stopwatch Y is also incorrect. S took no time at all, but is assigned a time of 1.
- Both deficiencies are corrected by stopwatch Z. It has nominal, ordinal and interval properties, but also has correct ratios and a correct zero point.

Summary of the Characteristics for Levels of Measurement (example 1)

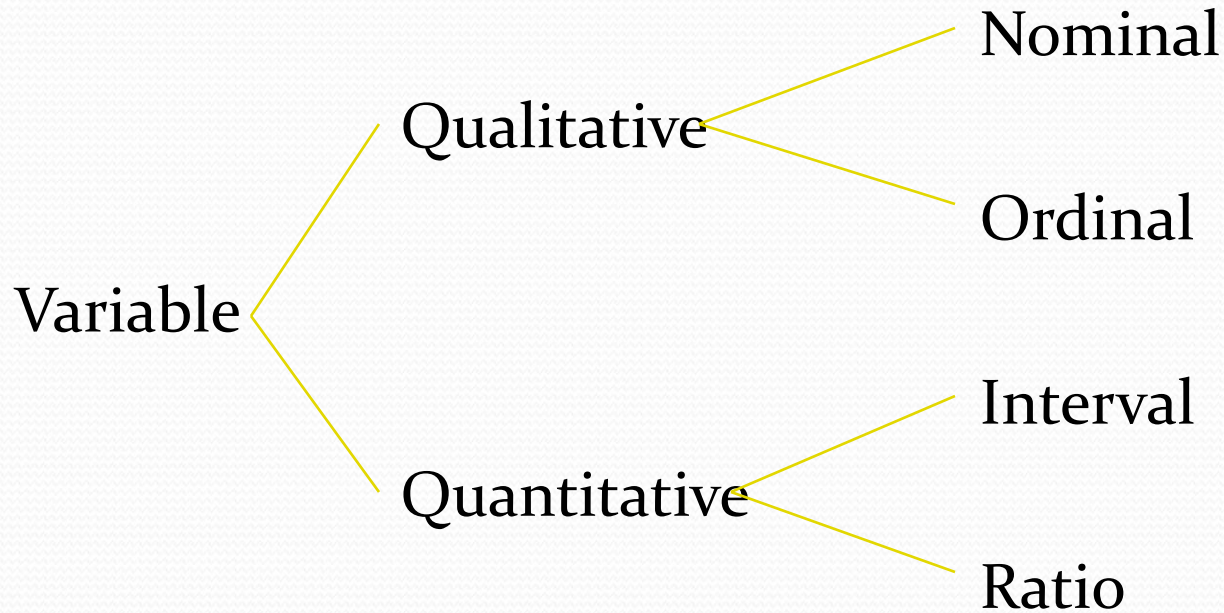


Other Examples

Find the levels of measurement of the following variables measured on college students:

- Residence: 1 = College dormitory, 2 = Off campus apartment, 3 = Parents' home, 4 = Other *Nominal*
- Whether or not the student's parents live in the same state as the student's college *Nominal*
- Class rank *Ordinal*
- Year graduated high school *Scale: Interval*
- Number of earned credits to date *Scale: Ratio*
- Height *Scale: Ratio*
- Family Income *Scale: Ratio*
- Number of siblings *Scale: Ratio*

Qualitative and quantitative variables may be further subdivided:



Data

- The measurements obtained in a research study are called the **data**.
- The goal of statistics is to help researchers organize and interpret the **data**.

